Операционные системы

Анализ файловой структуры UNIX. Команды для работы с файлами и каталогами

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Цели и задачи работы

Цель лабораторной работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

Задачи лабораторной работы

1 Выполнить приимеры

2 Выполнить дествия по работе с каталогами и файлами

3 Выполнить действия с правами доступа

4 Получить дополнительные сведения при помощи справки по командам.

Процесс выполнения лабораторной работы

Выполнение примеров

```
keisaev@keisaev:~$ touch abc1
keisaev@keisaev:~$ cp abc1 april
keisaev@keisaev:~$ cp abc1 may
keisaev@keisaev:~$ mkdir monthly
keisaev@keisaev:~$ cp april may monthly
keisaev@keisaev:~$ cp monthly/may monthly/june
keisaev@keisaev:~$ ls monthly
april  june  may
keisaev@keisaev:~$ mkdir monthly.00
keisaev@keisaev:~$ cp -r monthly monthly.00
keisaev@keisaev:~$ cp -r monthly.00 /tmp
keisaev@keisaev:~$
```

Рис. 1: Выполнение примеров

Выполнение примеров

```
keisaev@keisaev:~$
keisaev@keisaev:~$ mv april july
keisaev@keisaev:~$ mv july monthly.00
keisaev@keisaev:~$ ls monthly.00
july  monthly
keisaev@keisaev:~$ mv monthly.00 monthly.01
keisaev@keisaev:~$ mkdir reports
keisaev@keisaev:~$ mv monthly.01 reports
keisaev@keisaev:~$ mv reports/monthly.01 reports/monthly
keisaev@keisaev:~$
```

Рис. 2: Выполнение примеров

Выполнение примеров

```
keisaev@keisaev:~$
keisaev@keisaev:~$ touch may
keisaev@keisaev:~$ ls -l may
-rw-r--r--. 1 keisaev keisaev 0 сен 10 09:06 may
keisaev@keisaev:~$ chmod u+x may
keisaev@keisaev:~$ ls -l may
-rwxr--r--. 1 keisaev keisaev 0 сен 10 09:06 may
keisaev@keisaev:~$ chmod u-x may
keisaev@keisaev:~$ ls -l may
-rw-r--r--. 1 keisaev keisaev 0 сен 10 09:06 may
keisaev@keisaev:~$ chmod g-r,o-r monthly
keisaev@keisaev:~$ chmod g+w abc1
keisaev@keisaev:~$
```

Рис. 3: Выполнение примеров

Создание директорий и копирование файлов

```
keisaev@keisaev:~$
keisaev@keisaev:~$ cp /usr/include/linux/sysinfo.h ~
keisaev@keisaev:~$ mv sysinfo.h equipment
keisaev@keisaev:~$ mkdir ski.plases
keisaev@keisaev:~$ mv equipment ski.plases/
keisaev@keisaev:~$ mv ski.plases/equipment ski.plases/equiplist
keisaev@keisaev:~$ touch abc1
keisaev@keisaev:~$ cp abc1 ski.plases/equiplist2
keisaev@keisaev:~$ cd ski.plases/
keisaev@keisaev:~/ski.plases$ mkdir equipment
keisaev@keisaev:~/ski.plases$ mv equiplist equipment/
keisaev@keisaev:~/ski.plases$ mv equiplist2 equipment/
keisaev@keisaev:~/ski.plases$ cd
keisaev@keisaev:~$ mkdir newdir
keisaev@keisaev:~$ mv newdir ski.plases/
keisaev@keisaev:~$ mv ski.plases/newdir/ ski.plases/plans
keisaev@keisaev:~$
```

Рис. 4: Работа с каталогами

Работа с командой chmod

```
keisaev@keisaev:~$ mkdir australia play
keisaev@keisaev:~$ touch my_os feathers
keisaev@keisaev:~$
^[[200~chmod 744 australia/
^[[201~keisaev@keischmod 744 australia/tralia/
keisaev@keisaev:~$ chmod 711 play/
keisaev@keisaev:~$ chmod 544 my_os
keisaev@keisaev:~$ chmod 664 feathers
keisaev@keisaev:~$ ls -l
итого 0
-rw-rw-r--. 1 keisaev keisaev  0 сен 10 09:10  abc1
drwxr--r--. 1 keisaev keisaev  0 сен 10 09:13  australia
-rw-rw-r--. 1 keisaev keisaev  0 сен 10 09:13  feathers
drwxr-xr-x. 1 keisaev keisaev 74 сен  9 10:03  gt-extended
-rw-r--r--. 1 keisaev keisaev  0 сен 10 09:06  may
drwx--x--x. 1 keisaev keisaev 24 сен 10 08:59  monthly
-r-xr--r--. 1 keisaev keisaev  0 сен 10 09:13  my_os
drwx--x--x. 1 keisaev keisaev  0 сен 10 09:13  play
drwxr-xr-x. 1 keisaev keisaev 14 сен 10 09:06  reports
drwxr-xr-x. 1 keisaev keisaev 28 сен 10 09:11  ski.plases
drwxr-xr-x. 1 keisaev keisaev 10 сен  9 09:13  work
drwxr-xr-x. 1 keisaev keisaev  0 сен  9 08:33  Видео
drwxr-xr-x. 1 keisaev keisaev  0 сен  9 08:33  Документы
drwxr-xr-x. 1 keisaev keisaev 26 сен  9 09:16  Загрузки
drwxr-xr-x. 1 keisaev keisaev  0 сен  9 08:33  Изображения
drwxr-xr-x. 1 keisaev keisaev  0 сен  9 08:33  Музыка
drwxr-xr-x. 1 keisaev keisaev  0 сен  9 08:33  Общедоступные
drwxr-xr-x. 1 keisaev keisaev  0 сен  9 08:33 'Рабочий стол'
drwxr-xr-x. 1 keisaev keisaev  0 сен  9 08:33  Шаблоны
keisaev@keisaev:~$
```

Файл /etc/passwd

keisaev@keisaev:~ — less /etc/passwd

```
root:x:0:0:Super User:/root:/bin/bash
bin:x:1:1:bin:/bin:/usr/sbin/nologin
daemon:x:2:2:daemon:/sbin:/usr/sbin/nologin
adm:x:3:4:adm:/var/adm:/usr/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/usr/sbin/nologin
sync:x:5:0:sync:/sbin:/bin/sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
mail:x:8:12:mail:/var/spool/mail:/usr/sbin/nologin
operator:x:11:0:operator:/root:/usr/sbin/nologin
games:x:12:100:games:/usr/games:/usr/sbin/nologin
ftp:x:14:50:FTP User:/var/ftp:/usr/sbin/nologin
nobody:x:65534:65534:Kernel Overflow User:/:/usr/sbin/nologin
dbus:x:81:81:System Message Bus:/:/usr/sbin/nologin
apache:x:48:48:Apache:/usr/share/httpd:/sbin/nologin
tss:x:59:59:Account used for TPM access:/:/usr/sbin/nologin
avahi:x:70:70:Avahi mDNS/DNS-SD Stack:/var/run/avahi-daemon:/sbin/nologin
geoclue:x:999:999:User for geoclue:/var/lib/geoclue:/sbin/nologin
usbmuxd:x:113:113:usbmuxd user:/:/sbin/nologin
systemd-oom:x:998:998:systemd Userspace OOM Killer:/:/usr/sbin/nologin
qemu:x:107:107:qemu user:/:/sbin/nologin
polkitd:x:114:114:User for polkitd:/:/sbin/nologin
rtkit:x:172:172:RealtimeKit:/:/sbin/nologin
chrony:x:997:994:chrony system user:/var/lib/chrony:/sbin/nologin
dnsmasq:x:996:993:Dnsmasq DHCP and DNS server:/var/lib/dnsmasq:/usr/sbin/nologin
gluster:x:995:992:GlusterFS daemons:/run/gluster:/sbin/nologin
rpc:x:32:32:Rpcbind Daemon:/var/lib/rpcbind:/sbin/nologin
pipewire:x:994:991:PipeWire System Daemon:/run/pipewire:/usr/sbin/nologin
unbound:x:993:990:Unbound DNS resolver:/var/lib/unbound:/sbin/nologin
nm-openconnect:x:992:989:NetworkManager user for OpenConnect:/:/sbin/nologin
rpcuser:x:29:29:RPC Service User:/var/lib/nfs:/sbin/nologin
wsdd:x:991:988:Web Services Dynamic Discovery host daemon:/:/sbin/nologin
sssd:x:990:986:User for sssd:/run/sssd:/sbin/nologin
openvpn:x:989:985:OpenVPN:/etc/openvpn:/sbin/nologin
nm-openvpn:x:988:984:Default user for running openvpn spawned by NetworkManager:/:/sbin/nologin
flatpak:x:987:983:Flatpak system helper:/:/usr/sbin/nologin
colord:x:986:982:User for colord:/var/lib/colord:/sbin/nologin
abrt:x:173:173::/etc/abrt:/sbin/nologin
gdm:x:42:42:GNOME Display Manager:/var/lib/gdm:/usr/sbin/nologin
gnome-initial-setup:x:985:981::/run/gnome-initial-setup/:/sbin/nologin
vboxadd:x:984:1::/var/run/vboxadd:/sbin/nologin
/etc/passwd
```

Работа с файлами и правами доступа

```
keisaev@keisaev:~$ cp feathers file.old
keisaev@keisaev:~$ mv file.old play/
keisaev@keisaev:~$ mkdir fun
keisaev@keisaev:~$ cp -R play/ fun/
keisaev@keisaev:~$ mv fun/ play/games
keisaev@keisaev:~$ chmod u-r feathers
keisaev@keisaev:~$ cat feathers
cat: feathers: Отказано в доступе
keisaev@keisaev:~$ cp feathers feathers2
cp: невозможно открыть 'feathers' для чтения: Отказано в доступе
keisaev@keisaev:~$ chmod u+r feathers
keisaev@keisaev:~$ chmod u-x play/
keisaev@keisaev:~$ cd play/
bash: cd: play/: Отказано в доступе
keisaev@keisaev:~$ chmod +x play/
keisaev@keisaev:~$
```

Рис. 7: Работа с файлами и правами доступа

Справка по командам

keisaev@keisaev:~ — man mount

```
MOUNT(8)                                System Administration                                MOUNT(8)

NAME
       mount - mount a filesystem

SYNOPSIS
       mount [-h|-V]

       mount [-l] [-t fstype]

       mount -a [-fFnrsvw] [-t fstype] [-O optlist]

       mount [-fnrsvw] [-o options] device|mountpoint

       mount [-fnrsvw] [-t fstype] [-o options] device mountpoint

       mount --bind|--rbind|--move olddir newdir

       mount --make-[shared|slave|private|unbindable|rshared|rslave|rprivate|runbindable] mountpoint

DESCRIPTION
       All files accessible in a Unix system are arranged in one big tree, the file hierarchy, rooted at /. These files
       can be spread out over several devices. The mount command serves to attach the filesystem found on some device to
       the big file tree. Conversely, the umount(8) command will detach it again. The filesystem is used to control how
       data is stored on the device or provided in a virtual way by network or other services.

       The standard form of the mount command is:

          mount -t type device dir

       This tells the kernel to attach the filesystem found on device (which is of type type) at the directory dir. The
       option -t type is optional. The mount command is usually able to detect a filesystem. The root permissions are
       necessary to mount a filesystem by default. See section "Non-superuser mounts" below for more details. The
       previous contents (if any) and owner and mode of dir become invisible, and as long as this filesystem remains
       mounted, the pathname dir refers to the root of the filesystem on device.

       If only the directory or the device is given, for example:

          mount /dir

       then mount looks for a mountpoint (and if not found then for a device) in the /etc/fstab file. It's possible to
Manual page mount(8) line 1 (press h for help or q to quit)
```

Справка по командам

keisaev@keisaev:~ — man fsck

```
FSCK(8)                                System Administration                                FSCK(8)

NAME
       fsck - check and repair a Linux filesystem

SYNOPSIS
       fsck [-lsAVRTMNP] [-r [fd]] [-C [fd]] [-t fstype] [filesystem...] [--] [fs-specific-options]

DESCRIPTION
       fsck is used to check and optionally repair one or more Linux filesystems. filesystem can be a device name (e.g.,
       /dev/hdc1, /dev/sdb2), a mount point (e.g., /, /usr, /home), or a filesystem label or UUID specifier (e.g.,
       UUID=8868abf6-88c5-4a83-98b8-bfc24057f7bd or LABEL=root). Normally, the fsck program will try to handle
       filesystems on different physical disk drives in parallel to reduce the total amount of time needed to check all
       of them.

       If no filesystems are specified on the command line, and the -A option is not specified, fsck will default to
       checking filesystems in /etc/fstab serially. This is equivalent to the -As options.

       The exit status returned by fsck is the sum of the following conditions:

       0
           No errors

       1
           Filesystem errors corrected

       2
           System should be rebooted

       4
           Filesystem errors left uncorrected

       8
           Operational error

       16
           Usage or syntax error

       32
           Checking canceled by user request

Manual page fsck(8) line 1 (press h for help or q to quit)
```

Справка по командам

keisaev@keisaev:~ — man mkfs

```
MKFS(8)                                System Administration                                MKFS(8)

NAME
       mkfs - build a Linux filesystem

SYNOPSIS
       mkfs [options] [-t type] [fs-options] device [size]

DESCRIPTION
       This mkfs frontend is deprecated in favour of filesystem specific mkfs.<type> utils.

       mkfs is used to build a Linux filesystem on a device, usually a hard disk partition. The device argument is
       either the device name (e.g., /dev/hda1, /dev/sdb2), or a regular file that shall contain the filesystem. The
       size argument is the number of blocks to be used for the filesystem.

       The exit status returned by mkfs is 0 on success and 1 on failure.

       In actuality, mkfs is simply a front-end for the various filesystem builders (mkfs.fstype) available under Linux.
       The filesystem-specific builder is searched for via your PATH environment setting only. Please see the
       filesystem-specific builder manual pages for further details.

OPTIONS
       -t, --type type
           Specify the type of filesystem to be built. If not specified, the default filesystem type (currently ext2) is
           used.

       fs-options
           Filesystem-specific options to be passed to the real filesystem builder.

       -V, --verbose
           Produce verbose output, including all filesystem-specific commands that are executed. Specifying this option
           more than once inhibits execution of any filesystem-specific commands. This is really only useful for
           testing.

       -h, --help
           Display help text and exit.

       -V, --version
           Print version and exit. (Option -V will display version information only when it is the only parameter,
           otherwise it will work as --verbose.)
Manual page mkfs(8) line 1 (press h for help or q to quit)
```

Справка по командам

keisaev@keisaev:~ — man kill

```
KILL(1)                                User Commands                                KILL(1)

NAME
       kill - terminate a process

SYNOPSIS
       kill  [-signal|-s signal|-p]  [-q value] [-a] [--timeout milliseconds signal] [--] pid|name...

       kill -l [number] | -L

DESCRIPTION
       The command kill sends the specified signal to the specified processes or process groups.

       If no signal is specified, the TERM signal is sent. The default action for this signal is to terminate the
       process. This signal should be used in preference to the KILL signal (number 9), since a process may install a
       handler for the TERM signal in order to perform clean-up steps before terminating in an orderly fashion. If a
       process does not terminate after a TERM signal has been sent, then the KILL signal may be used; be aware that the
       latter signal cannot be caught, and so does not give the target process the opportunity to perform any clean-up
       before terminating.

       Most modern shells have a builtin kill command, with a usage rather similar to that of the command described
       here. The --all, --pid, and --queue options, and the possibility to specify processes by command name, are local
       extensions.

       If signal is 0, then no actual signal is sent, but error checking is still performed.

ARGUMENTS
       The list of processes to be signaled can be a mixture of names and PIDs.

       pid
           Each pid can be expressed in one of the following ways:

           n
               where n is larger than 0. The process with PID n is signaled.

           0
               All processes in the current process group are signaled.

           -1
               All processes with a PID larger than 1 are signaled.

Manual page kill(1) line 1 (press h for help or q to quit)
```

Выводы по проделанной работе

Вывод

В ходе данной работы мы ознакомились с файловой системой Linux, её структурой, именами и содержанием каталогов. Научились совершать базовые операции с файлами, управлять правами их доступа для пользователя и групп. Ознакомились с Анализом файловой системы. А также получили базовые навыки по проверке использования диска и обслуживанию файловой системы.